

Notes: Bernard, Eaton, Jensen, and Kortum (AER, 2003)

Plants and Productivity in International Trade

Contents

1	Big picture	3
2	Incremental contribution of the paper	3
3	Data and measurement	4
3.1	Plant-level data	4
3.2	Aggregate trade and production data	4
3.3	Plant-level outcomes of interest	4
4	Model and empirical specifications	5
4.1	Demand	5
4.2	Technology and delivered costs	5
4.3	Bertrand pricing and markups	5
4.4	Trade shares	5
4.5	Plant-level productivity	6
4.6	Parameterization and moments	6
4.7	Productivity decomposition in counterfactuals	6
5	Identification and empirical design	6
6	Tables and figures	8
6.1	Table 1: Plant-Level Export Facts + Figure 1: Industry Exporting Intensity	8
6.2	Figure 2A: Ratio of Plant Labor Productivity to Overall Mean + Figure 2B: Ratio to 4-Digit Industry Mean	8
6.3	Table 2: Plant-Level Productivity Facts	9
6.4	Table 3: Aggregate Trade Data	9
6.5	Table 4: Export Facts, Simulated vs. Actual + Table 5: Counterfactuals	11
6.6	Table 6: Plant-Level Transitions after Lower Geographic Barriers + Table 7: Transitions after Higher U.S. Relative Wage	12
7	Short summary	12
8	Conclusion	13
8.1	Core conclusion	13
8.2	Mechanism	13
8.3	Incremental contribution revisited	13
8.4	Policy and counterfactual implications	13
8.5	Limitations and extensions	13

8.6 Takeaway	13
------------------------	----

1 Big picture

- **Question.** How can a general-equilibrium trade model explain the plant-level facts that exporters are rare, much larger and more productive than nonexporters, yet typically export only a small share of output?
- **Core answer.** Combine Ricardian productivity heterogeneity, iceberg trade barriers, and Bertrand competition with variable markups. More efficient plants can beat rivals in more markets, charge higher markups, and appear more productive in revenue-based productivity measures.
- **Empirical target.** The model is disciplined by bilateral manufacturing trade among the United States and 46 major trading partners, plus plant-level facts from the 1992 U.S. Census of Manufactures.
- **Key facts to match.** Only 21 percent of U.S. manufacturing plants export; most exporters sell less than 10 percent of output abroad; exporters are much larger; exporters have a 33 percent unconditional labor-productivity advantage; large productivity dispersion remains within industries and factor-intensity bins.
- **Quantitative fit.** With estimated parameters $\theta = 3.60$ and $\sigma = 3.79$, the model matches the exporter productivity and size premia and captures the skewness of export intensity, although it predicts too many plants export.
- **Counterfactuals.** A 5 percent fall in geographic barriers raises U.S. manufacturing productivity by 4.7 percent, lowers the number of plants by 3.3 percent, but reduces manufacturing employment by only 1.3 percent. A 10 percent rise in the U.S. relative wage raises productivity by 4.2 percent but reduces employment by 13 percent.
- **Economic takeaway.** Even when exports are a small share of plant revenue, the international economy has large effects on which plants survive, which plants export, how employment reallocates, and how aggregate productivity changes.

2 Incremental contribution of the paper

- **From aggregate trade to plants.** Earlier Ricardian trade models explained bilateral trade shares and aggregate productivity, while the new empirical exporter literature documented plant-level heterogeneity. This paper connects the two in one quantitative framework.
- **A pre-Melitz route to exporter selection.** Unlike fixed-export-cost models, exporter selection here comes from head-to-head competition and variable geographic barriers. Plants export only when their delivered cost advantage beats foreign competitors after trade costs.
- **Variable markups as the bridge from efficiency to measured productivity.** Under perfect competition or common-markup monopolistic competition, revenue productivity would not reveal underlying efficiency. Bertrand competition makes more efficient producers tend to have higher markups, so measured productivity varies across plants.
- **Micro moments discipline macro trade parameters.** The model uses exporter productivity and size premia to pin down key parameters, then asks whether it also matches export participation, export intensity, and productivity dispersion.
- **Plant-level counterfactuals from a general-equilibrium model.** The paper shows how aggregate shocks, such as globalization or dollar appreciation, produce heterogeneous plant transitions: entry, exit, switching into exporting, switching out of exporting, and reallocating employment.
- **Interpretive contribution.** The paper reframes trade liberalization: aggregate productivity gains are not only price-index effects but also reallocation from low-productivity domestic plants to high-productivity exporters.

3 Data and measurement

3.1 Plant-level data

- The plant-level facts come from the 1992 U.S. Census of Manufactures in the Longitudinal Research Database.
- The sample contains over 200,000 manufacturing plants, excluding very small plants not mailed a census form.
- The data include shipments, employment, wages, value added, capital stock, ownership, and reported exports.
- A plant's export measure includes direct exports and products shipped to exporters or wholesalers for export, but the paper notes undercounting relative to aggregate export data.
- Labor productivity is measured as value added per worker, with exporters and nonexporters compared both unconditionally and within industry/factor-intensity categories.

Interpretation: The micro data provide the facts that standard trade models could not explain: most plants do not export, exporters export little as a share of output, and productivity dispersion remains large within industries.

3.2 Aggregate trade and production data

- The model uses 47 countries or regions: the United States plus the largest U.S. manufacturing export destinations, with some consolidations to address entrepot trade and Caribbean Basin apparel trade anomalies.
- Bilateral manufacturing trade flows come from Feenstra, Lipsey, and Bowen data.
- Gross manufacturing production comes from OECD, UNIDO, or, when necessary, World Bank value-added data scaled to gross production.
- The paper computes home purchases as gross production minus total exports, and absorption as the sum of domestic purchases plus imports.

Interpretation: The aggregate trade data identify relative technology, geography, and market size conditions that determine which countries supply which markets.

3.3 Plant-level outcomes of interest

- **Export participation:** whether a plant exports anything.
- **Export intensity:** exports as a share of plant shipments among exporters.
- **Productivity dispersion:** standard deviation of log value added per worker.
- **Exporter productivity advantage:** average log productivity of exporters minus nonexporters.
- **Plant size advantage:** difference between exporters and nonexporters in shipments or domestic shipments.
- **Counterfactual transition status:** exit, domestic-only production, or exporting after a shock.

Interpretation: These objects allow the authors to ask whether a macro trade model can generate the joint distribution of size, productivity, and exporting at the plant level.

4 Model and empirical specifications

4.1 Demand

Consumers in country n have CES demand over a continuum of goods $j \in [0, 1]$. Expenditure on good j is

$$X_n(j) = X_n \left(\frac{p_n(j)}{p_n} \right)^{1-\sigma}, \quad \sigma > 1. \quad (1)$$

Interpretation: Demand is elastic. A plant that can charge a lower delivered price can capture a larger share of a destination market.

4.2 Technology and delivered costs

For each good, country, and producer rank, efficiency is random. The delivered unit cost from country i to destination n is

$$C_{kni}(j) = \frac{w_i d_{ni}}{Z_{ki}(j)}, \quad d_{ii} = 1, \quad d_{ni} \leq d_{nk} d_{ki}. \quad (2)$$

Efficiencies follow a Frechet distribution,

$$F_i(z) = \Pr[Z_i \leq z] = \exp(-T_i z^{-\theta}). \quad (3)$$

Interpretation: T_i shifts the country's technology level and θ governs dispersion. Lower θ means more heterogeneity and therefore stronger plant-level selection.

4.3 Bertrand pricing and markups

Let $C_{1n}(j)$ and $C_{2n}(j)$ denote the lowest and second-lowest delivered costs for good j in market n . The winning supplier prices at

$$p_n(j) = M_n(j) C_{1n}(j), \quad M_n(j) = \min \left\{ \frac{C_{2n}(j)}{C_{1n}(j)}, \frac{\sigma}{\sigma - 1} \right\}. \quad (4)$$

Interpretation: The markup depends on the cost gap to the nearest rival. Efficient plants tend to have bigger cost gaps, higher markups, and higher measured revenue productivity.

4.4 Trade shares

The probability that country i supplies country n is the standard Eaton-Kortum share:

$$\pi_{ni} = \frac{T_i (w_i d_{ni})^{-\theta}}{\Phi_n}, \quad \Phi_n = \sum_k T_k (w_k d_{nk})^{-\theta}. \quad (5)$$

Interpretation: Aggregate bilateral trade shares discipline technology, wages, and trade barriers. Plant-level outcomes are generated underneath these aggregate shares.

4.5 Plant-level productivity

The model's value-added-per-worker measure is proportional to

$$\frac{v(j)}{w_i} = \frac{X(j) - (1 - \beta)I(j)}{w_i L(j)}, \quad (6)$$

where $X(j)$ is total revenue, $I(j)$ is total input cost, and β is labor's share of variable cost.

Interpretation: Because $X(j)/I(j)$ is a composite markup, revenue productivity varies because markups vary. This is why Bertrand competition is central.

4.6 Parameterization and moments

- The paper sets the intermediate share using the 1992 U.S. manufacturing intermediate share of revenues, 0.63.
- It chooses θ and σ to match two U.S. plant moments: the productivity advantage of exporters and the size advantage of exporters.
- The estimated values are $\theta = 3.60$ and $\sigma = 3.79$.

Interpretation: A lower θ raises underlying efficiency dispersion; a higher σ makes demand more responsive to price differences and affects the size advantage.

4.7 Productivity decomposition in counterfactuals

The counterfactual productivity change is decomposed into continuing-plant productivity changes, reallocation among continuers, covariance terms, entry, and exit:

$$q' - q = \sum_{j \in \Omega_c} s(j)[q'(j) - q(j)] + \sum_{j \in \Omega_c} [s'(j) - s(j)][q(j) - q] \quad (7)$$

$$+ \sum_{j \in \Omega_c} [s'(j) - s(j)][q'(j) - q(j)] + \sum_{j \in \Omega_n} s'(j)[q'(j) - q] + \sum_{j \in \Omega_x} s(j)[q - q(j)]. \quad (8)$$

Interpretation: This decomposition shows whether productivity gains come from within-plant changes, reallocation toward better plants, entry, or exit of weak plants.

5 Identification and empirical design

- **Structural identification.** The model is not estimated as a causal regression; instead, it is disciplined by matching aggregate bilateral trade flows and plant-level exporter moments.
- **Why bilateral trade matters.** Whether a U.S. plant exports to France depends not only on U.S. productivity but also on the delivered costs of competitors from Germany, Japan, Canada, and many other suppliers.
- **Why plant-level moments matter.** Aggregate trade shares alone do not identify the degree of producer heterogeneity needed to explain exporter productivity and size premia.
- **Counterfactual logic.** Once parameters are fixed, the model simulates production efficiencies for many goods, solves baseline and counterfactual allocations, and tracks plant outcomes.

- **Key assumptions.** The model abstracts from fixed export costs, multi-product plants, industry heterogeneity, dynamics, capital adjustment, and internal trade barriers.
- **Credibility.** The model is credible as a disciplined quantitative mechanism linking macro trade and micro plant heterogeneity, but the counterfactual magnitudes should be read as suggestive rather than definitive.

6 Tables and figures

6.1 Table 1: Plant-Level Export Facts + Figure 1: Industry Exporting Intensity

Read:

- Table 1 shows that 79 percent of plants do not export and only 21 percent export anything.
- Among exporters, 66 percent export only 0–10 percent of shipments and 82 percent export less than 20 percent.
- Figure 1 shows that exporters are spread across industries: in most industries, some but not all plants export.
- No single industry bin explains export participation; the export decision is largely within-industry.

Economic message: These facts motivate a plant-level theory of trade. A model in which entire industries export or do not export cannot explain the coexistence of exporters and nonexporters within narrowly defined industries.

TABLE 1—PLANT-LEVEL EXPORT FACTS	
Export status	Percentage of all plants
No exports	79
Some exports	21
Export intensity of exporters (percent)	Percentage of exporting plants
0 to 10	66
10 to 20	16
20 to 30	7.7
30 to 40	4.4
40 to 50	2.4
50 to 60	1.5
60 to 70	1.0
70 to 80	0.6
80 to 90	0.5
90 to 100	0.7

Note: The statistics are calculated from all plants in the 1992 Census of Manufactures.

(a) Table 1: Plant-Level Export Facts

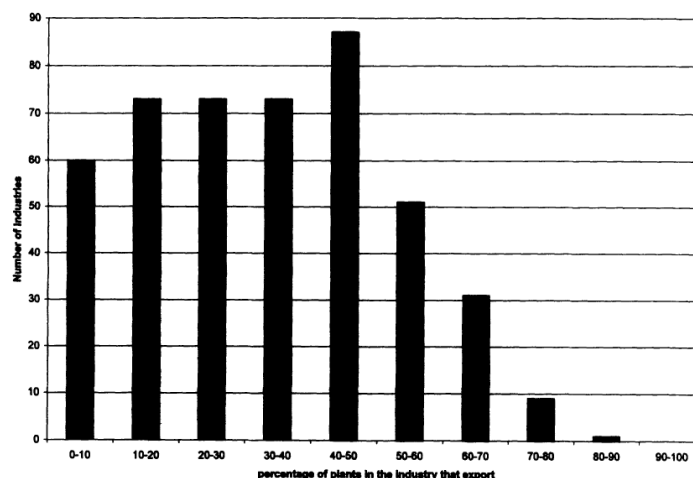


FIGURE 1. INDUSTRY EXPORTING INTENSITY

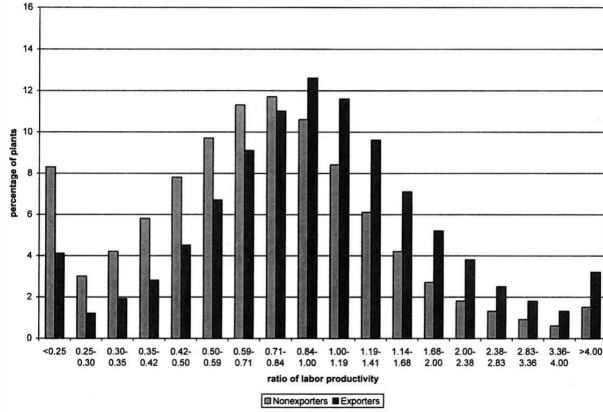
(b) Figure 1: Industry Exporting Intensity

6.2 Figure 2A: Ratio of Plant Labor Productivity to Overall Mean + Figure 2B: Ratio to 4-Digit Industry Mean

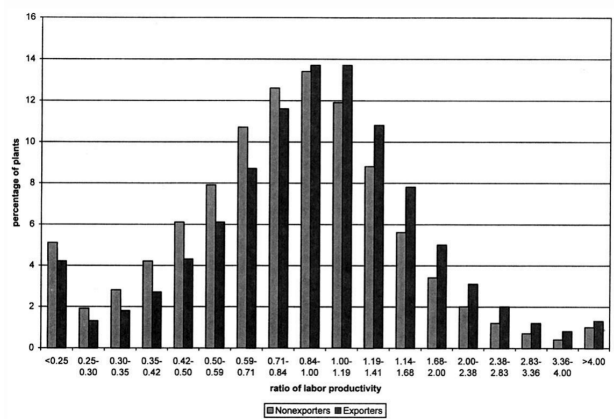
Read:

- Figure 2A shows broad productivity dispersion across plants, with exporters shifted to the right of nonexporters.
- Figure 2B repeats the comparison after normalizing by the 4-digit industry mean. The distribution tightens only modestly.
- Exporters remain more productive even within industries.

Economic message: Industry composition is not enough. A model must explain within-industry productivity heterogeneity and why productive plants export.



(a) Figure 2A: Productivity Relative to Overall Mean



(b) Figure 2B: Productivity Relative to Industry Mean

6.3 Table 2: Plant-Level Productivity Facts

Read:

- The standard deviation of log productivity is 0.75 unconditionally and 0.66 within 4-digit industries.
- Exporters have a 33 percent unconditional productivity advantage.
- Controlling for 4-digit industry reduces the advantage to 15 percent; controlling for both industry and factor-intensity bins leaves a 9–11 percent advantage.

Economic message: Even after accounting for industry, capital intensity, and skill intensity, exporters remain more productive. This supports a model with idiosyncratic plant efficiency, not just sectoral comparative advantage.

FIGURE 2B. RATIO OF PLANT LABOR PRODUCTIVITY TO 4-DIGIT INDUSTRY MEAN

Productivity measure (value added per worker)	Variability (standard deviation of log productivity)	Advantage of exporters (exporter less nonexporter average log productivity, percent)
Unconditional	0.75	33
Within 4-digit industries	0.66	15
Within capital-intensity bins	0.67	20
Within production labor-share bins	0.73	25
Within industries (capital bins)	0.60	9
Within industries (production labor bins)	0.64	11

Notes: The statistics are calculated from all plants in the 1992 Census of Manufactures. The “within” measures subtract the mean value of log productivity for each category. There are 450 4-digit industries, 500 capital-intensity bins (based on total assets per worker), 500 production labor-share bins (based on payments to production workers as a share of total labor cost). When appearing within industries there are 10 capital-intensity bins or 10 production labor-share bins.

Table 2: Plant-Level Productivity Facts

6.4 Table 3: Aggregate Trade Data

Read:

- Table 3 lists the 47 countries/regions used to discipline the aggregate trade side of the model.
- The U.S. has a high home market share, but market shares vary sharply by partner: Canada and Mexico/Caribbean have much larger U.S. shares than Europe or Japan.

- The table also reports imports from and exports to the rest of the world, showing that undercounting from omitting smaller countries is limited.

Economic message: The model is not calibrated only to U.S. plants. It embeds U.S. plants in a global system of bilateral competition, because a U.S. exporter must beat rival suppliers from many countries.

TABLE 3—AGGREGATE TRADE DATA

No.	Country	Data source	U.S. exports (\$ million)	U.S. percent market share	Imports from ROW (percent of total)	Exports to ROW (percent of total)
1	Arab Emirates	W	1,590	7.9	5.3	36.4
2	Argentina	U	3,498	3.8	2.8	11.9
3	Australia	O	8,570	6.2	2.9	5.9
4	Austria	O	1,785	1.5	5.9	12.5
5	Belgium and Luxembourg	O, U	6,264	4.3	4.2	3.4
6	Brazil	U	5,932	2.9	3.4	7.2
7	Canada	O	83,400	32.9	0.7	0.7
8	Chile	U	2,441	8.8	1.6	2.7
9	China and Hong Kong	U, U	16,200	3.3	1.7	6.2
10	Colombia	U	3,098	12.5	1.8	5.4
11	Denmark	O	1,403	2.5	4.7	7.5
12	Ecuador	U	1,035	14.9	1.2	1.1
13	Egypt	U	1,665	7.1	6.5	22.7
14	Finland	O	914	1.8	2.7	4.9
15	France	O	16,700	2.4	4.0	10.6
16	Germany (unified)	O	23,000	1.8	6.6	7.6
17	Greece	O	804	2.0	4.1	11.9
18	India	U	1,624	1.3	8.4	9.7
19	Indonesia	U	2,846	5.1	0.9	3.9
20	Ireland	U	2,771	8.4	1.8	1.9
21	Israel	U	3,251	10.0	1.5	8.7
22	Italy	O	8,124	1.2	6.4	9.0
23	Japan	O	42,100	1.6	2.5	4.3
24	Korea (South)	O	14,100	5.5	1.4	6.9
25	Kuwait	U	1,471	15.5	3.8	25.2
26	Mexico and Caribbean	O, U, W, U, U	43,700	19.8	2.4	7.0
27	Netherlands	O	9,362	5.1	2.2	4.6
28	New Zealand	O	1,526	7.3	1.1	8.4
29	Nigeria	U	1,012	6.5	1.2	1.0
30	Norway	O	1,779	3.2	3.0	8.5
31	Paraguay	W	807	11.5	0.9	3.9
32	Peru	U	858	5.8	2.8	2.1
33	Philippines	U	1,667	4.9	1.1	1.4
34	Portugal	U	807	1.3	1.7	8.0
35	Saudi Arabia	W	7,145	14.4	2.1	8.1
36	Singapore and Malaysia	U, U	15,000	14.1	1.5	6.4
37	Spain	O	5,717	1.9	2.6	7.6
38	Sweden	O	3,403	3.1	3.5	5.0
39	Switzerland	U	4,222	3.2	2.7	5.2
40	South Africa	U	2,106	3.0	1.5	6.1
41	Taiwan	U	14,000	8.1	0.6	1.3
42	Thailand	U	4,094	3.7	3.1	7.3
43	Turkey	U	2,186	2.4	6.0	16.9
44	United Kingdom	O	22,600	3.8	2.0	11.4
45	United States	O	2,520,300	85.0	1.8	3.5
46	USSR (former)	U	2,181	0.4	8.6	9.3
47	Venezuela	U	6,390	17.2	1.4	10.2

Table 3: Aggregate Trade Data

6.5 Table 4: Export Facts, Simulated vs. Actual + Table 5: Counterfactuals

Read:

- Table 4 shows that the model predicts 51 percent of plants export, versus 21 percent in the Census.
- Conditional on exporting, the model matches the skewness of export intensity: it predicts 76 percent of exporters sell 0–10 percent abroad, compared with 66 percent in the data.
- Table 5 shows that a 5 percent decline in geographic barriers raises productivity by 4.7 percent, while a 10 percent rise in the U.S. relative wage raises productivity by 4.2 percent.
- Lower barriers increase U.S. exports by 11 percent and imports by 17 percent; a higher U.S. relative wage cuts U.S. exports by 14 percent and manufacturing employment by 13 percent.

Economic message: The model fits some export facts better than others. It overpredicts export participation but captures the key fact that most exporters remain primarily domestic. Counterfactuals show that aggregate productivity gains can accompany substantial plant churn.

Export status	Percentage of all plants	
	Simulated	Actual
No exports	49	79
Some exports	51	21

Export intensity of exporters (percent)	Percentage of exporting plants	
	Simulated	Actual
0 to 10	76	66
10 to 20	19	16
20 to 30	4.2	7.7
30 to 40	0.0	4.4
40 to 50	0.0	2.4
50 to 60	0.0	1.5
60 to 70	0.0	1.0
70 to 80	0.0	0.6
80 to 90	0.0	0.5
90 to 100	0.0	0.7

(a) Table 4: Export Facts

Statistics for U.S. producers	Counterfactual experiment	
	5-percent lower barriers	10-percent higher U.S. relative wage
Productivity decomposition (percent change):		
Aggregate	4.7	4.2
Entrants	0.0	0.0
Exiters	0.8	0.8
Reallocation, continuing plants	0.2	−0.2
Covariance, continuing plants	−0.1	−0.1
Gains, continuing plants	3.9	3.7
Plant exit and entry:		
Number of plants (percent change)	−3.3	−3.1
Relative productivity of exiters (percent)	45	47
Employment share (percent), prior	1.5	1.4
Employment:		
Total employment in manufacturing (percent change)	−1.3	−13
Job creation (percent)	1.5	0.3
Job destruction (percent)	2.8	13
International trade:		
U.S. exports (percent change)	11	−14
U.S. imports (percent change)	17	10
U.S. absorption (percent change)	−1.9	−6.0
World trade (percent change)	15	1.4

(b) Table 5: Counterfactuals

6.6 Table 6: Plant-Level Transitions after Lower Geographic Barriers + Table 7: Transitions after Higher U.S. Relative Wage

Read:

- Table 6 shows the micro transitions behind globalization. Among initially domestic plants, 6.7 percent exit and 5.2 percent begin exporting. Exit is concentrated among low-productivity domestic plants.
- High-productivity domestic plants are much more likely to enter export markets after barriers fall: 12 percent begin exporting in the highest quartile.
- Table 7 shows transitions after a U.S. relative wage increase. Domestic plants face exit risk, while exporters mostly continue to export but may retreat toward domestic production.
- In the lowest productivity quartile, 15 percent of initially domestic plants exit after the wage shock; among low-quartile exporters, 2.6 percent exit and 18 percent become domestic-only.

Economic message: Aggregate trade shocks have heterogeneous effects. Low-productivity domestic plants are exposed to import competition, while high-productivity plants are the ones that can expand into export markets or survive wage pressure.

	Before geographic barriers fall	After geographic barriers fall		
	Percent of plants	Exit (percent)	Domestic (percent)	Export (percent)
All plants:				
Domestic	49	6.7	88	5.2
Export	51	0.1	0.9	99
By productivity quartile:				
Lowest quartile				
Domestic	77	17	80	2.9
Export	23	0.6	1.5	98
Second quartile				
Domestic	66	0.1	95	4.6
Export	34	0.0	1.6	98
Third quartile				
Domestic	30	0	93	7.2
Export	70	0	0.9	99
Highest quartile				
Domestic	21	0	88	12
Export	79	0	0.4	99

	Before appreciation of the U.S. wage	After appreciation of the U.S. wage		
	Percent of plants	Exit (percent)	Domestic (percent)	Export (percent)
All plants:				
Domestic	49	6.1	94	0.0
Export	51	0.3	10	90
By productivity quartile:				
Lowest quartile				
Domestic	77	15	85	0.0
Export	23	2.6	18	79
Second quartile				
Domestic	66	0.0	100	0.0
Export	34	0.0	18	82
Third quartile				
Domestic	31	0	100	0
Export	69	0	11	89
Highest quartile				
Domestic	21	0	100	0
Export	79	0	3.9	96

(a) Table 6: Lower Geographic Barriers

(b) Table 7: Higher U.S. Relative Wage

7 Short summary

- The paper builds a quantitative model of international trade that speaks directly to plant-level facts.
- It combines Ricardian productivity heterogeneity, iceberg trade barriers, and Bertrand competition.
- Bertrand competition creates variable markups, which make more efficient plants appear more productive in revenue-based measures.
- The model explains why exporters are more productive and larger, why most plants do not export, and why exporters export only a small share of output.
- Plant-level heterogeneity remains important even within industries and factor-intensity groups.
- The model uses bilateral trade data for 47 countries/regions and 1992 U.S. Census plant data.
- Parameter estimates are $\theta = 3.60$ and $\sigma = 3.79$.
- The model overpredicts the share of plants that export but captures the skewed export intensity distribution.

- Globalization raises productivity by reallocating activity away from low-productivity domestic plants and toward high-productivity exporters.
- A stronger U.S. wage reduces employment much more than a symmetric fall in trade barriers, even though both shocks raise average productivity.

8 Conclusion

8.1 Core conclusion

Bernard, Eaton, Jensen, and Kortum show that a parsimonious Ricardian model with imperfect competition can reconcile aggregate trade flows with rich plant-level facts. Plants differ in efficiency, face destination-specific trade barriers, and compete head-to-head with rivals. These ingredients generate export selection, exporter size premia, and exporter productivity premia.

8.2 Mechanism

The core mechanism is cost-rank competition. A plant exports to a market only if its delivered cost is lower than rivals' delivered costs after trade barriers. The same efficiency that helps a plant win markets also tends to give it a larger cost gap over its nearest rival, which raises its markup and measured productivity. Thus, export status, size, and measured productivity all move together even without fixed export costs.

8.3 Incremental contribution revisited

The paper's lasting contribution is to bridge two literatures that had been largely separate: aggregate Ricardian trade theory and plant-level evidence on exporters. It shows that bilateral trade shares and micro producer heterogeneity can be disciplined in a unified model. This is conceptually distinct from representative-firm trade theory and from fixed-export-cost models because trade participation is generated by variable trade costs and head-to-head global competition.

8.4 Policy and counterfactual implications

The counterfactuals show why plant-level heterogeneity matters for trade policy. Lower trade barriers can raise average productivity while shutting down low-productivity plants and reallocating labor toward high-productivity exporters. A real appreciation or wage shock can raise measured productivity by forcing weak plants out, but it can also generate large employment losses. Aggregate productivity gains therefore need not imply benign distributional or employment effects.

8.5 Limitations and extensions

The model abstracts from dynamics, fixed costs, multiproduct firms, industry heterogeneity, capital adjustment, and internal trade barriers. These omissions are important because observed plant behavior may reflect sunk export costs, learning, financing constraints, or multi-product reallocations. Still, the model is a foundational step because it demonstrates that a disciplined general-equilibrium trade model can generate meaningful plant-level outcomes.

8.6 Takeaway

The main lesson is that international trade reshapes production through selection and reallocation among heterogeneous plants. Even when most exporters sell mostly at home, foreign markets and foreign competitors determine which plants are viable, which ones export, and how aggregate productivity evolves.